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Identification of Behavioral Function in Public Schools

And a Clarification of Terms

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Abstract

The discipline-related component of IDEA requires that schools conduct a Functional Behavioral Assessment (FBA) when a student's behavior disrupts the educational environment and / or results in suspension from school. Applied behavior analysts often make a distinction between the terms *functional assessment / functional behavioral assessment* and *functional analysis* yet there exists no consensus on how that distinction should be made. A relevant review of the literature was conducted to identify research articles using functional analysis or functional assessment methodology in public school settings in an effort to identify the specific procedures employed by each. Results of the review support the existence of a discrepancy between proposed and actual school-based assessment models, as well as other claims regarding functional assessment research. We address the problem of distinguishing between the terms *assessment* and *analysis* as they relate to procedures employed to determine behavioral function of students exhibiting aberrant behavior. A clarification of terms is proposed.

While the primary purpose of schools remains educating children, safety and discipline have become an integral part of our nation's goals for education (Crone & Horner, 1999). Traditionally, schools have relied on punitive approaches toward behavior management to include detention and suspension (Crone & Horner, 1999; Kern, Childs, Dunlap, Clarke, & Falk, 1994), often with administrators involved in dealing with problem behavior (Crone & Horner, 1999). As identified by Drasgow and Yell (2001), arranging school environments that are safe and conducive to learning, while protecting the rights of students with disabilities was a Congressional goal for the reauthorization of the Individuals with Disabilities Education Act (IDEA) in 1997. The discipline-related component of IDEA requires that schools conduct a Functional Behavioral Assessment (FBA) when a student's behavior disrupts the educational environment, impedes a student's learning, or that of his peers, and / or when the student's behavior results in a suspension of more than 10 days (Miller, Tansy, & Hughes, 1998).

FBA is the process of gathering information that provides insight into the behavioral function of learned aberrant behaviors (i.e., positive, negative, and automatic reinforcement). The process reveals patterns illustrating the contextual influences that set the occasion for and maintain occurrences of behavior. Its purpose is to guide the development of individualized behavior support plans by determining why a student engages in a particular behavior(s) (Lohrmann-O'Rourke, Knoster, & Llewellyn, 1999).

FBA leads to an intervention plan based on information provided through the assessment process. Behavior analytic literature suggests that there is a direct relationship between the identification of the function of the behavior and effective treatment (Carr & Durand, 1985; Iwata, Dorsey, Slifer, Bauman, &Richman, 1982/1994;

Iwata, Pace, Cowdery, & Miltenberger, 1994; Mace, Lalli, & Lalli, 1991). As identified by Watson, Ray, Turner, and Logan (1999), a key issue in school-based intervention is determining the type of assessment data needed to develop an effective intervention. The approach taken by applied researchers to identify the environmental variables maintaining aberrant behavior has been to isolate the events responsible for the occurrence and / or non-occurrence of the behavior. This helps to identify a causal relationship between environmental events and changes in behavior (Iwata, Vollmer, & Zarcone, 1990), and in the behavior analytic literature is referred to as functional analysis methodology (Iwata et al., 1982 / 1994). Although FBA is not yet defined by the U.S. Department of Education, it is assumed that Congress intended the term be interpreted in a manner consistent with that provided in the professional literature (Drasgow &Yell, 2001; Dunlap & Kincaid, 2001).

FBA has its roots in functional analysis methodology such as that provided in the research conducted by Iwata, et al. (1982/1994). While FBA refers to a range of procedures that can be used to identify antecedents and consequences associated with aberrant behavior, often it can be differentiated from functional analysis by the presence or absence of the condition / phase including experimental manipulation of the variables hypothesized to maintain problem behavior (experimental analysis of behavior) (Ervin, Radford, Bertsch, Piper, Ehrhardt, & Poling, 2001; Gresham, Watson, & Skinner, 2001; Wehby, Symons & Shores, 1995).

It is this experimental analysis of behavior coupled with indirect and descriptive procedures that constitutes best practice in assessing behavioral function (Ervin et al., 2001; Sterling-Turner, Robinson, & Wilczynski, 2001). More specifically, indirect methods typically include a review of student records, teacher or caregiver interviews, and/ or the use of rating scales and checklists (Gresham, Watson, & Skinner, 2001). Indirect methods alone however, are considered insufficient in determining behavioral function (Sterling-Turner, Robinson, & Wilczynski, 2001).

Descriptive methods, naturalistic observations, or antecedent-behavior-consequence (A-B-C) analyses are conducted in the individual's natural environment without manipulating the variables that are associated with aberrant behavior. Both descriptive and experimental analyses are considered direct methods (as opposed to indirect described above) in that they involve direct observation of the behavior in question vs. a reliance on third-party reporting as in the interview process. Similar to experimental analyses, the primary objective of descriptive analysis is the development of data-based hypotheses regarding the function of aberrant behavior (i.e., the functional relation between aberrant behavior and naturally occurring environmental events). Other valuable information, however, can be obtained via direct observation of naturally occurring antecedent and consequent events. Examples of such information include the schedule of reinforcement available for appropriate or aberrant behavior, the skills required for successful functioning, individual styles of interaction, and the variables involved in response efficiency. Information obtained via descriptive methods provides an empirical basis for the formulation of hypotheses regarding behavioral function (Lalli & Goh, 1993). These hypothesized functions are then typically evaluated through an experimental analysis; however, descriptive procedures alone have been found to lead to appropriate and effective treatments (Lalli, Browder, Mace, & Brown, 1993; Kates-McElrath, Flor, & Lalli, 2001; Sterling-Turner, Robinson, & Wilczynski, 2001). The literature on functional analysis is primarily clinical in nature with few experimental analyses occurring in natural settings while enlisting school personnel in the process (Ervin et al., 2001; Ervin, DuPaul, Kern, & Friman, 1998).

Several issues have evolved from the reauthorization of IDEA that concern school personnel and behavior analysts alike. First, there exists no consensus regarding how an FBA should be defined including the specific procedures that should constitute an FBA (Asmus, Vollmer, & Borrero, 2002; Ervin, Ehrhardt, & Poling, 2001; March & Horner, 2002; Sterling-Turner, Robinson, & Wilczynski, 2001). Second, behavior analysts often make a distinction between the terms *functional assessment / functional behavioral assessment* and *functional analysis* (Gresham, Watson, & Skinner 2001), yet there exists no consensus on how that distinction should be made (Dunlap & Kincaid, 2001) or whether, in fact, the distinction is necessary at all (Gresham, Watson, & Skinner, 2001; Gresham, McIntyre, Olson-Tinker, Dolstra, McLaughlin, & Van, 2003). Third, there seem to be conflicting ideas regarding the purpose of experimentally analyzing the environmental conditions suspected of maintaining aberrant behavior, whether that be for purposes of hypothesis testing (Kern et al., 1994) or hypothesis-generating (Gable, 1996). Finally, although IDEA indicates training for school personnel involved in the use of functional assessment is necessary, it does not specify the minimal conditions for this training (Ervin, Ehrhardt, & Poling, 2001). In their 1999 article, Crone & Horner identify a discrepancy between legislative mandates for FBA and the skills, resources, and training available in school settings to effectively integrate these mandates.

Although there are several important concerns raised in the previous paragraph, addressing all of them extends beyond the scope of this paper. Hence, the purpose of this paper is to address the first two concerns relating to the specific procedures that define FBA, as well as the often interchangeable use of the terms *assessment* and *analysis*. A relevant review of the literature was conducted to identify research articles using functional analysis or functional assessment methodology in public school settings in an effort to identify the specific procedures employed by each. We address the problem of distinguishing between the terms *assessment* and *analysis* as they relate to procedures employed to determine behavioral function of students exhibiting aberrant behavior. As a result, a clarification of terms is proposed.

Method

A review of the relevant literature on the application of behavioral assessment methodology was conducted by three educational psychology graduate students. The literature review was conducted using the Temple University online library (*PsycINFO*) with the following key terms: functional assessment; functional analysis; functional behavioral assessment; descriptive analysis; naturalistic observations; consultant model; and / or teacher-collected data. Requirements for consideration for selection included research whose primary setting was a public school system, either special or general education classrooms, extending from kindergarten through high-school age (school-age). The included research must have contained a specified methodology that was implemented with resulting data reported. Articles that were primarily theoretical, conceptual, or presented only a proposed model were not considered for selection. Initially, articles published between 1995 and 2005 were targeted. Due to lack of research identified in public school settings within that time frame, the search was extended backward by three years (1992). Articles published prior to 1992 were not included.

Of the articles selected, the primary areas of interest were the following: the journal in which the article was published, the setting in which the research was conducted (elementary, middle, or high school, special or general education), the specific participant population, the procedures identified for functional assessment / analysis (indirect, descriptive, experimental methods), the format identified for hypothesis testing (in vivo, analog, etc.), and the model for data collection (consultant, teacher, student / self).

Results and Discussion

The results of the literature review are summarized in Table 1. The articles meeting criteria for inclusion in the study from 1992 – 2002 totaled 11 articles. They are listed in Table 2. The *Journal of Applied Behavior Analysis* provided the majority of articles involving functional assessment / analysis research in school-based settings (64%). The results of the review (91%) supported the claim that the majority of functional analysis research has focused on children with developmental disabilities (Gresham, et al., 2004; Hanley, Iwata, & McCord, 2003; Larson & Maag, 1998). Furthermore, the setting for functional analysis research in schools has often been special education classrooms as opposed to general education or inclusion settings (Hanley et al., 2003). Our findings were also consistent with this claim in that the majority of the research was conducted in special education settings (54%), with few analyses conducted in general education or inclusion settings (9%) (other settings were unclear or unspecified).

With regard to functional assessment / analysis procedures, the majority of analyses were conducted using indirect and direct (descriptive and / or experimental) methods (72%) and were not all published in the *Journal of Applied Behavior Analysis*. Of these articles however, only 36% employed indirect, descriptive, and experimental methods in their assessment process while another 36% employed only indirect and descriptive methods (no experimental analysis). No articles reported using indirect procedures alone, which supports the claim that indirect procedures are generally considered insufficient in determining behavioral function without extending the analysis to direct observations with or without experimental manipulations (Sterling-Turner et al., 2001). Nine percent of the articles each reported using descriptive methods alone, descriptive with experimental methods (no indirect phase), or indirect with experimental (no descriptive phase) methods.

On the issue of hypothesis testing, 45% of the articles reported testing hypotheses directly within the classroom setting vs. 27 % using analog conditions). Articles reporting hypothesis testing in analog conditions were published only in the *Journal of Applied Behavior Analysis* (100%). Eighteen percent of the articles reported testing hypotheses *in vivo* (in classroom) only manipulated the variables that were suspected to play an explicit role in behavioral function. Another 18% reported testing hypotheses through an evaluation of treatment effects by monitoring the effectiveness of the intervention without further testing.

The means for data collection in all of the articles was that of the consultant model (100%) in which a person employed privately, or employed directly but providing services to the classroom in an itinerant manner, guided the assessment and intervention process. Fifty-four percent however, did incorporate classroom teachers into the process while only one article (9%) incorporated the student in their own analysis. Again, these findings are consistent with claims in the literature that the majority of studies have been conducted under controlled conditions with assessment and intervention guided by the investigators rather that school personnel (Ervin et al., 2001; Quinn, Gable, Fox, Rutherford, Van Acker, & Conroy, 2001).

There reportedly exists a substantial increase in the literature concerning FBA's and school settings in the past 20 years; however, widespread variations in the assessment process make it difficult to identify the essential components (Ervin et al., 2001). Based on the articles we selected for inclusion, there exists little agreement on which procedures are necessary and sufficient for the FBA process in school settings. In addition, 54% of the articles referred to their procedures as *functional assessment* while 45% referred to them as *functional analysis.* All studies using the term *functional analysis* included an experimental analysis of behavior; however, the experimental phase was included for some of the studies whose authors referred to their procedures as *functional assessment.*

General consensus seems to be that the term *functional assessment* is an overarching term that does not necessarily involve the experimental analysis phase owned by the term *functional analysis* (Ervin et al., 2001; Gable, 1996; Gresham, Watson, & Skinner, 2001; Tobin, 1994; Wehby, Symons & Shores, 1995). The increased popularity of the term *functional assessment* perhaps results from the reauthorization of

IDEA and the subsequent translation of behavioral theory into practice in school settings. *Functional assessment* however can be found in the behavioral literature pertaining to school-based settings prior to 1997 (Karsh, Repp, Dahlquist, & Monk, 1995; Taylor & Romanczyk, 1994; Sasso et al., 1992). Further definition of these procedures is clearly required to address its effective implementation in school settings (Ervin et al., 2001).

Several researchers have attempted to successfully extrapolate functional assessment / analysis methodology to school settings (Dunlap & Kincaid, 2001; Kern et al., 1994; Lalli et al., 1993; O'Neill, Horner, Albin, Storey, & Sprague, 1997; Northup, Wacker, Berg, Kelly, Sasso, & DeRaad, 1994; Repp, Felce, & Barton, 1988; Repp & Karsh, 1994; Sasso et al., 1992; Taylor & Romanczyk, 1994). A significant number of more recent articles however, offer a conceptual view and / or proposed model for integrating functional assessment / analysis methodology into school settings (Asmus, Vollmer, & Borrero, 2002; Crone & Horner, 1999; Drasgow &Yell, 2001; Dunlap & Kincaid, 2001; Ervin, Ehrhardt, & Poling, 2001; Gable, 1996; Gresham et al., 2004; Gresham, Watson, & Skinner, 2001; Larson & Maag, 1998; McComas & Mace, 2000; Quinn et al., 2001; Roberts, 2002**;** Sugai & Horner, 1999; Tobin, 1994). Of these conceptual articles, only one was published in the *Journal of Applied Behavior Analysis* while three were published by the *School Psychology Review*.

Sugai and Horner (http://www.pbis.org/ news/v1i1/index.htm) remind us that schools should be commended for their efforts in adopting a function-based, theoretically sound, and empirically validated approach toward behavioral support. However, IDEA has forced functional assessment methodology into the hands of practitioners who, while skilled in their own professions, may lack the specific theoretical training, expertise, and resources for effective implementation of behavior analytic methodology. In this case, there is the potential of the procedures and methodology being misapplied (Sugai & Horner, http://www.pbis.org/ news/v1i1/index.htm). As pointed out by Asmus, Vollmer, & Borrero (2002), the field of applied behavior analysis has the means and opportunity to assist schools in the development of effective and appropriate assessment procedures. Yet, it would appear that this area of need has been neglected by applied behavior analysts given the existing discrepancies noted previously, that little research is conducted in general education settings, the ongoing issues and problems pertaining to staff training and availability of resources, and the notion that functional behavioral assessment is quickly becoming an integral part of the practice of school psychologists (Ervin, Ehrhardt, & Poling, 2001).

Future Research and Implications

Given the significant number of articles offering conceptual views on integrating functional assessment / analysis into school settings (Asmus, Vollmer, & Borrero, 2002), and the lack of agreement regarding the defining procedures (Ervin, et al., 2001), further investigation is warranted. The experimental investigation of the use of functional assessment-related procedures in classroom settings should be complimented by investigations into training protocols for teachers and other school personnel who might be called upon to conduct or participate in these assessments. In addition, future research should not only be disseminated in primarily behavior-based journals (e.g.., *Journal of Applied Behavior Analysis*), but in journals that are more commonly read by classroom personnel (i.e., *Exceptional Children*) and school-based psychologists (i.e., *School*

Psychology Review) in an effort to disseminate best practice recommendations and identify guidelines that can serve as resources for training individuals who are using functional assessment procedures.

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